Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Observation

Meaning in my words How would you explain observation to a friend?	**How can you tell?** The timer reads 2 s as the cart passes the 4 m mark. What did you actually see?
Example or sketch What two labels would you put on a sketch of that moment?	**Non-example and why** Someone writes “It will reach 8 m.” Has the camera seen that yet?

Model

Meaning in my words How would you explain model to a friend?	**How can you tell?** A cart moves 2 m each second. What are we assuming if we extend that pattern?
Example or sketch Where would your model put the cart one second after it passes 6 m going right?	**Non-example and why** The cart hits a cushion. Would you still extend the same pattern? Why?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Prediction

Meaning in my words How would you explain prediction to a friend?	**How can you tell?** Which part of a forecast could we check with the next camera image?
Example or sketch A tracker reads 0, 2, then 4 km a minute apart. Where would you look a minute later?	**Non-example and why** Someone calls yesterday's recorded position a prediction. What would you call it?

Testable question

Meaning in my words How would you explain testable question to a friend?	**How can you tell?** What would you measure to answer "Does this cart speed up on the ramp?"
Example or sketch What question could two timed pictures of the same cart help you answer?	**Non-example and why** "Which cart looks coolest?" What would you change so the class can test it?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw three observed positions and one predicted position. Build a key that separates the two.

Choose zero, direction, and scale.



My marks mean

One more exposure

01

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Separate a sighting from a prediction.

CHOOSE ONE JOB TOGETHER

TELESCOPE SCHEDULER ☐

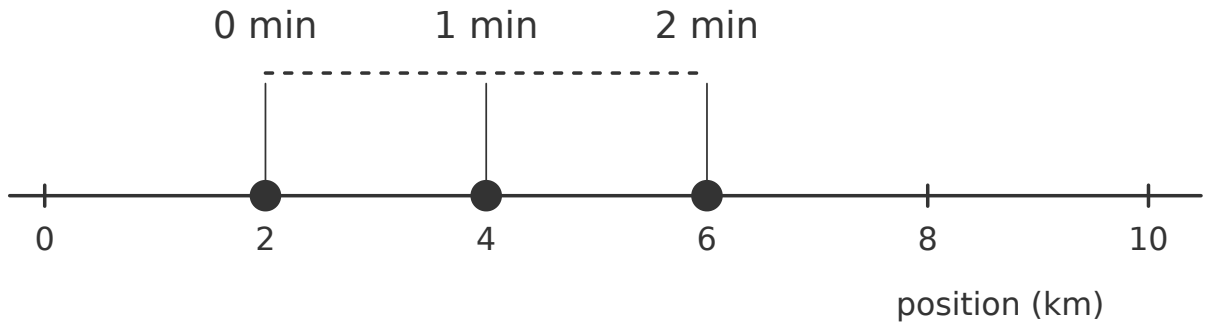
Mark where to aim at minute **3**. Tell the operator which dots were measured and which dot you predicted.

SCIENCE BROADCASTER ☐

Give viewers a two-sentence update. Point to a measured dot, then say where you expect the next one.

Point to a clue. Keep your first marks; add the update using the color key.

Three telescope fixes are at **2**, **4** and **6 km** at minutes **0**, **1** and **2**. The night crew has one more exposure.



01 OUR FIRST CALL

Where should the camera look next? Use the timed dots in your call.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new sighting fit your prediction? Tell the next crew what to check.

Switch speakers. Which dot did we measure, and which did we predict?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks